What we found and what is next

Preliminary findings • Controlled, noise-free regression • GD vs L-BFGS

MAIN FINDING

As κ grows, GD needs far more updates.

746

GD updates

5

L-BFGS iterations

At κ = 100 • Same starting weights and stopping threshold

NEXT STEPS

1. **Repeat from new starts**
2. **Measure gradient calls and runtime**
3. **Try mini-batch SGD or feature scaling**

PRESENTATION PLAN

Original: 7:00 • Rehearsal target: 6:30

Suggested split: 1:10 + 1:20 + 1:20 + 1:40 + 1:00

Preliminary results • All six reported runs meet ||gradient|| < 10^{-6} • Fewer iterations ≠ lower total cost.